



Midterm Examinations

1st Semester, S.Y. 2024 – 2025
Transcribed by: Aerith Claude Catap
Instructor: Mrs. Kawinya T. Manzón

HAU School of Computing
First Year | Bachelor of Science in
Computer Science

Syllabus:

- 2.1. Technology as a Way of Revealing
- 2.2. Human Flourishing in Progress and De-development
- 2.3. The Good Life

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Technology as a Way of Revealing

Who is Martin Heidegger?

- A widely acknowledged German **philosopher** of the 20th century whose focus is on ontology of the study of **'being'** or **dasein** (in German).
- Continental tradition of philosophy
- Joined the Nazi Party (NSDAP)
- Stern opposition to positivism and technological world domination
- Philosophical works are often described as complicated.
- "The essence of technology is by no means technological."

The Question Concerning Technology

- A work by **Martin Heidegger**, in which the author discusses the essence of technology.
- Heidegger originally published the text in 1954 in his book, **Vorträge und Aufsätze**.
- Heidegger initially developed the themes in the text in the lecture **"The Framework" (Das Gestell)**, first presented on **December 1, 1949**, in **Bremen**.
 - "The Framework" was presented as the second of four lectures, collectively called **"Insight into what is."** The other lectures were titled "The Thing" (*Das Ding*), "The Danger" (*Die Gefahr*), and "The Turning" (*Die Kehre*).

The Essence of Technology

- The question concerning technology is asked, as Heidegger notes, "so as to prepare a free relationship to it". The relationship will be free "if it opens our **human existence (Dasein)** to the **essence of technology**".
- Thus, questioning uncovers technology in its (true) essence as it is; enabling it to be "experienced within its own bounds" by seeking "the true by way of the correct".
- Heidegger begins the question by noting that "We ask the question concerning technology when we ask **what it is**". This follows an ancient doctrine which states that "the essence of a thing is considered to be what the thing is".
- He further states that "**Technology is a means to an end [and] a human activity**". This means humans set goals and use technology to achieve them.
 - If technology is a means to a human end, this conception can therefore be "called the instrumental and anthropological definition of technology".

- This leads to the question, "what is the instrumental itself?" In other words, it asks us to explore the nature of means and ends and to consider, "within what do such things as means and end belong?".
 - To simplify this context, technology is referred to as a **"means to an end"** – it is something that humans use to achieve a specific goal or outcome.
- A means can be seen as that through and by which an end is effected. It is that "whereby something is effected and thus attained". In essence, it can be seen as a cause, for "Whatever has an effect as its consequence is called a cause". But an **end is also a cause** to the extent that it determines the kind of means to be used to actualize it.
 - A "means" can also be considered as a **cause** because it brings out the outcome you're aiming for.
 - The ends (goal) also act as a cause because it determines which means will be used. In essence, when we talk about using means to achieve ends, we are also talking about **causality** – how causes lead to effects.
 - Hence, when we use **technology** as a means to an end, we understand that technology serves as a tool to help us achieve specific goals. The technology we choose is influenced by the desired ends we want to reach.
- As noted, "The end in keeping with which the kind of means to be used is determined is also considered a cause". This conceptualization of instrumentality as means and ends leads the question further into causality, suggesting that "wherever ends are pursued and means are employed, wherever instrumentality reigns, there reigns causality".
 - In this context, **instrumentality** is about how tools and methods (means) are used to achieve specific goals (ends).
 - This relationship between means and ends leads to **causality**, where both are interconnected, and their interaction helps to explain how actions lead to results. This will be further elaborated below.
- To question **causality**, Heidegger starts from what "for centuries philosophy has taught" regarding the traditional "four causes.". These are traditionally enumerated as:
 1. **Causa materialis** – the material, the matter out of which" something is made
 2. **Cause formalis** – the form, the shape into which the material enters
 3. **Causa finalis** – the end, in relation to which [the thing] required is determined as to its form and matter"
 4. **Causa efficiens** – which brings about the effect that is the finished [thing]".
- Heidegger concludes that "what technology is, when represented as a means, **discloses itself when we trace instrumentality back to fourfold causality**". To explain this, Heidegger uses the example of a silver chalice. Each element works together to create the chalice in a different manner:
- Thus four ways of owing hold sway in the sacrificial vessel that lies ready before us. They differ from one another, yet they belong together. ... The four ways of being responsible bring something into appearance. They let it come forth into **presencing**. They set it free to that place and so start it on its way, namely into its complete arrival.



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Technology as Poiesis

- When the abovementioned four elements work together to create something into appearance, it is called **bringing-forth**.
- This bringing-forth comes from the Greek *poiesis*, which "**brings out of concealment into unconcealment**". This process of revealing can be represented by the Greek word *aletheia*, which in English is translated as "truth". This truth has everything to do with the essence of technology because **technology is a means of revealing the truth**.
- Modern technology, however, differs from *poiesis*. Heidegger suggests that this difference stems from the fact that modern technology "is based on modern physics as an exact science"
- The revealing of modern technology, therefore, is not bringing-forth, but rather **challenging-forth**. To exemplify this, Heidegger draws on the Rhine River as an example of how our modern technology can change a cultural symbol.
- To further his discussion of modern technology, Heidegger introduces the notion of **standing-reserve**. He suggests that modern technology turns people into resources that are simply available and waiting to be used. For example, a forester is like a standing-reserve for the paper and print industries, waiting to fulfill their needs.

Modern Technology as Enframing

- Heidegger once again returns to discuss the essence of modern technology to name it Gestell, which he defines primarily as a sort of enframing:
- **Enframing** means the gathering together of that setting-upon that sets upon man, i.e., challenges him forth, to reveal the real, in the mode of ordering, as standing-reserve. Enframing means that way of revealing that holds sway in the essence of modern technology and that it is itself not technological.
 - In simpler terms, **enframing** is a concept where modern technology urges us to see everything, including ourselves, as resources or tools to be used. It shapes how we perceive the world by making everything appear as a resource waiting to be used, but this way of seeing is not itself a technology.
- Once he has discussed enframing, Heidegger highlights the threat of technology. As he states, this threat "does not come in the first instance from the potentially lethal machines and apparatus of technology". Rather, the **threat is the essence** because "the rule of enframing threatens man with the possibility that it could be denied to him to enter into a more original revealing and hence to experience the call of a more primal truth".
 - Here, Heidegger explains that the real danger of technology does not come from dangerous or lethal machines. **Rather, the threat lies in the essence of technology** – as technology's way of framing everything (especially our environment) as resources might prevent us from experiencing deeper, more fundamental truths.
- This is because challenging-forth conceals the process of bringing-forth, which means that truth itself is concealed and no longer unrevealed. Unless humanity makes an effort to re-orient itself, it will not be able to find revealing and truth.

- It is at this point that Heidegger has encountered a paradox. – Humanity must be able to navigate the dangerous orientation of enframing because it is in this dangerous orientation that we find the potential to be rescued.
 - While technology's way of framing everything can be dangerous, it also holds the potential for deeper understanding and truth.
 - To further elaborate on this, Heidegger returns to his discussion of essence. Ultimately, he concludes that "the essence of technology is in a lofty sense ambiguous" and that "such ambiguity points to the mystery of all revealing, i.e., of truth".
- The question concerning technology, Heidegger concludes, is one "concerning the constellation in which revealing and concealing, in which the coming to presence of the truth comes to pass". In other words, **it is finding truth**.
- Heidegger presents **art** as a way to navigate this constellation, this paradox, because the artist, or the poet as Heidegger suggests, views the world as it is and as it reveals itself.
 - He suggests that art, especially through poets and artists, can help us navigate this paradox by showing us both how the world is and how it reveals itself.
- **For questioning is the piety of thought.**

In conclusion, Heidegger's "The Question Concerning Technology" can thus aptly be described as a comprehensive attempt to interrogate the idea of technology in order to gain an understanding of the essence of the thing, rather than merely understanding it as an instrument or a means.

Human Flourishing in Progress and De-development

- This section presents Jason Hickel's development framework focused on de-development. Below is an excerpt from one of his works, "*Forget 'developing' poor countries, it's time to 'de-develop' rich countries*"
- As a departure from traditional frameworks of growth and development, Hickel's concept of de-development is discussed as an alternative to narrowing the gap between rich and poor countries. Thus, taking off from this alternative framework, the section critiques human flourishing vis-a-vis progress in science and technology.
- Growth has been the main object of development for the past 70 years, despite the fact that it's not working. Since 1980, the global economy has grown by 380%, but the number of people living in poverty on less than \$5 (£3.20) a day has increased by more than **1.1 billion**. That's 17 times the population of Britain.
- **Orthodox economists** insist that all we need is yet more growth. More progressive types tell us that we need to shift some of the yields of growth from the richer segments of the population to the poorer ones, evening things out a bit.
 - **Neither approach is adequate**; because even at current levels of average global consumption, we're overshooting our planet's bio-capacity **by more than 50% each year**.
- In other words, growth isn't an option any more – we've already grown too much. Scientists are now telling us that we're blowing past planetary boundaries at breakneck speed. And the hard truth is



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that this global crisis is due almost entirely to **overconsumption in rich countries.**

- **Instead of pushing poor countries to 'catch up' with rich ones, we should be getting rich countries to 'catch down'**
- Right now, our planet only has enough resources for each of us to consume 1.8 “global hectares” annually – a standardized unit that measures resource use and waste.
- What does this mean for our theory of development?
 - **Economist Peter Edward** argues that instead of pushing poorer countries to “catch up” with rich ones, we should be thinking of ways to get rich countries to “catch down” to more appropriate levels of development.
 - We should look at societies where people live long and happy lives at relatively low levels of income and consumption not as basket cases that need to be developed towards western models, but as exemplars of efficient living.
- How much do we really need to live long and happy lives?
 - In the US, life expectancy is 79 years and GDP per capita is \$53,000. But many countries have achieved similar life expectancy with a mere fraction of this income.
 - Cuba has a comparable life expectancy to the US and one of the highest literacy rates in the world with GDP per capita of only \$6,000 and consumption of only 1.9 hectares – right at the threshold of ecological sustainability. Similar claims can be made of Peru, Ecuador, Honduras, Nicaragua and Tunisia.
- Yes, some of the excess income and consumption we see in the rich world yields improvements in quality of life that are not captured by life expectancy, or even literacy rates. But even if we look at measures of overall happiness and wellbeing in addition to life expectancy, a number of low- and middle-income countries rank highly. Costa Rica manages to sustain one of the highest happiness indicators and life expectancies in the world with a per capita income one-fourth that of the US.
- In light of this, perhaps we should regard such countries **not as underdeveloped, but rather as appropriately developed.** And maybe we need to start calling on rich countries to justify their excesses.
- The idea of “de-developing” rich countries might prove to be a strong rallying cry in the global south, but it will be tricky to sell to westerners. Tricky, but not impossible.
- According to recent consumer research, **70% of people** in middle- and high-income countries believe overconsumption is putting our planet and society at risk. A similar majority also believe we should strive to buy and own less, and that doing so would not compromise our happiness. People sense there is something wrong with the dominant model of economic progress and they are hungry for an alternative narrative.
- The problem is that the pundits promoting this kind of transition are using the wrong language.
 - They use terms such as de-growth, zero growth or – worst of all – de-development, which are technically accurate but off-putting for anyone who's not already on board. Such terms are repulsive because they run against the deepest frames we use to think about human progress, and, indeed, the purpose of life itself.

- The idea of “**steady-state**” **economics** is a step in the right direction and is growing in popularity, but it still doesn't get the framing right. We need to reorient ourselves toward a positive future, a truer form of progress.
 - One that is geared toward **quality instead of quantity.** One that is more sophisticated than just accumulating ever increasing amounts of stuff, which doesn't make anyone happier anyway. What is certain is that GDP as a measure is not going to get us there and we need to get rid of it.
- The west has its own tradition of reflection on the good life and it's time we revive it. **Robert and Edward Skidelsky** take us down this road in his book **How Much is Enough?** where they lay out the possibility of interventions such as banning advertising, a shorter working week and a basic income, all of which would improve our lives while reducing consumption.
- **Either we slow down voluntarily or climate change will do it for us.** We can't go on ignoring the laws of nature.

The Good Life

Aristotle

- Born in **Greece**, Aristotle lived from **384 BC to 322 BC.**
- The influence of Aristotle's work on the physical sciences spread far and wide, offering well thought out theory and reasoning that would prevail for many years.
- The main point of Aristotle in living a good life is **virtue ethics** which is mainly based on the rational account of a good human life and identifies good human life with virtuous life and **virtue is conceived as human excellence.**
- **The good life therefore is the life of excellence.** For Aristotle, this is the main purpose of human existence.
- This highest good is happiness or **eudaimonia** (which translates to living well).
- Aristotle thought of **happiness as a central purpose of human life.** He also believed that our happiness was dependent on ourselves rather than others. “Happiness is the meaning and the purpose of life: the whole aim and end of human existence.”
 - According to Aristotle, happiness consists in achieving, through the course of a whole lifetime, all the goods — **health, wealth, knowledge, friends, etc.** — that lead to the perfection of human nature and to the enrichment of human life. This requires us to make choices, some of which may be very difficult.
- According to Aristotle, there are three types of friendships:
 - Those based on **utility**, where people associate for their mutual usefulness. These relationships are the most common.
 - Those based on **pleasure or delight**, and
 - Those grounded in **virtue.**
- The moral theory of Aristotle, like that of Plato, **focuses on virtue**, recommending the virtuous way of life by its relation to happiness.
- In subsequent books, excellent activity of the soul is tied to the moral virtues and to the virtue of “practical wisdom” — excellence in thinking and deciding about how to behave.



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- Moral virtues are exemplified by **courage**, **temperance**, and **liberality**; the key intellectual virtues are **wisdom**, which governs ethical behavior, and **understanding**, which is expressed in scientific endeavor and contemplation. For example, regarding what are the most important virtues, Aristotle proposed the following nine: **wisdom**; **prudence**; **justice**; **fortitude**; **courage**; **liberality**; **magnificence**; **magnanimity**; **temperance**.

Aristotle, "What is the Life of Excellence?"

- For human beings, eudaimonia is activity of the soul in accordance with **arete** (excellence, virtue, or what something is good for").
- Happiness, the end of life, that to which all things aim, **is activity in accordance with reason** (the arete or peculiar excellence of a person).
- **Generosity** as a moral virtue achieves a mean between wastefulness and greed.
- According to Aristotle, the intellectual virtues include:
 1. **Episteme** – scientific knowledge
 2. **Techne** – artistic or technical knowledge
 3. **Nous** – intuitive reason
 4. **Phronesis** – practical wisdom
 5. **Sophia** – philosophic wisdom